Problem Set 1

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Advanced Methods in Applied Statistics
Feb - Mar 2026

Photo by Howard Jackman

Format

- The submission is:
  - A write-up as a PDF document, which includes any plots, diagrams, tables, pictures, and explanations
  - In a separate "file", submit all code used to derive the results
    - Tarball, zipped directory, lots of individual files w/ self-explanatory titles, etc.
  - Include any original data files or how the data was accessed
    - If you use a internet scraping tool, note the date when you retrieved the data
    - If you can save the data to a file, do so and submit the data file. There is no need to change the format, e.g. HTML, XML, JSON, etc.
  - **Cover page (0.5 pts)**
    - On the cover page include your *name*, the *date of submission*, a facial *image of yourself*, the *name of the course*, and the *title of the submission*
  - Submit the assignment (both the write-up and code) via Absalon

Example Cover Page (0.5 pts.)

Problem Set #1

David “Jason” Koskinen (XDN365)

Advanced Methods in Applied Statistics
Day Month Year

Software and Data Handling

- As a precursor to doing computer aided statistics, the first problem set will focus on data handling, parsing text, writing code, and simple presentation
- Problems will focus on USA college basketball statistics from the 2014 and 2009 Ken Pomeroy Basketball pages at https://kenpom.com/
  - The content is unimportant and was chosen due to some *interesting* features
  - There are many, many, many different ways to use the data from the 2014 and 2009 webpages, and I will note that some webscraping software and tools are blocked by the KenPom.com website. For those who experience web scraping blocking, I have received permission from Ken Pomeroy to host a static and minimal version of the relevant webpages at:
    - https://www.nbi.dk/~koskinen/Teaching/AdvancedMethodsInAppliedStatistics2026/data/2014KenPomeroy.html
    - https://www.nbi.dk/~koskinen/Teaching/AdvancedMethodsInAppliedStatistics2026/data/2009KenPomeroy.html

Time

- This will be *potentially* time-consuming
  - Took me 4 hours to do the first time and 2 hours to redo
  - Could be done in ~20 minutes if you've already mastered: regular expressions, HTML/XML parser, class declaration, plotting commands, etc…

Assignment Overview

- The goal of the first assignment is to assess how well students can load, parse, minimally analyze, and plot data
  - A plotting and data-throughput exercise
  - But, there are some interesting data features
- Words of advice for the following problem set
  - Don’t be overly reliant on spreadsheets
  - Don’t assume that the input data (or format) is stable between years for Problems 2 and 3
- There are some known (at least by me) ambiguities in the problems. If you come across what you perceive is an ambiguity, detail it in your write-up.

Problem 1 (2.5 pts.)

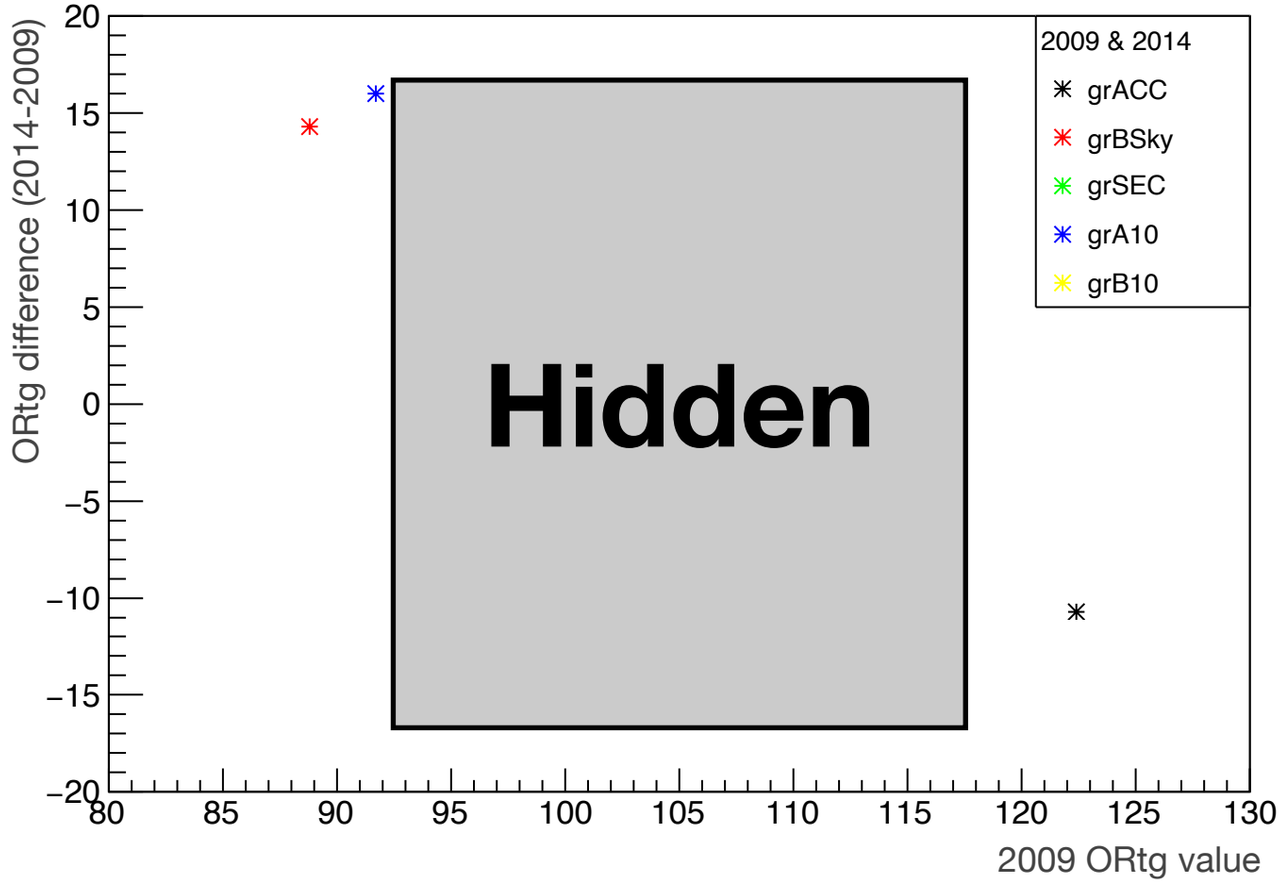
- Take the 2014 Ken Pomeroy data related to NCAA College Basketball analytics from http://kenpom.com/index.php?y=2014
- On a single plot, produce histograms of:
  - The Adjusted Defense Efficiency "DRtg" for all the teams in the 5 conferences (ACC, SEC, B10, BSky, and A10)
    - The Adjusted Defensive Efficiency (DRtg) has a range of approximately 85-125
  - Have different colors for each conference and add a legend
  - The plot must be visually understandable.
    - E.g. keep bin widths constant for all the data if you plot a histogram.

Problem 2 (4 pts.)

- Take the 2014 and 2009 Ken Pomeroy data related to NCAA College Basketball analytics
- Calculate the difference in "ORtg" for all the teams in the 5 conferences from Exercise 1:
  - 2014 minus 2009 as a function of the 2009 ORtg value
    - The Adjusted Offensive Efficiency (ORtg) has a range of approximately 85-125
  - Plot the data as a graph with a data point for each team entry being the same conference color as for the previous plot in Exercise 1
- Calculate the median and mean of the difference in "ORtg" between 2009 and 2014:
  - For each of the 5 conferences (there should be a median and a mean for each conference)
  - For all teams that were not in the 5 conferences (there should a single mean, and a single median when considering all of the teams NOT in the 5 conferences covered above).

Problem 2 cont. (Example)

- Calculate the difference in "ORtg" for all the teams in the 5 conferences from Exercise 1:
  - Between 2014 and 2009 as a function of the 2009 ORtg value
  - Plot the data as a graph with a data point for each team entry being the same conference color as for the previous histograms



*Be mindful that this plot is an example and is not guaranteed to be accurate

**Each team should have its own data point. There are many more points beneath the 'Hidden' box

Problem 3 (3pts.)

- Take the 2014 and 2009 Ken Pomeroy data related to NCAA College Basketball analytics
- Redo Problems 1 and 2, while now adding the "BE" conference to the previous list of 5 conferences
  - For those who have written robust code for the earlier problems, this should be fast and easy
  - It is likely to be much harder for those who…
    - Parse some data in "by hand"
    - Only wrote code that requires the exact data format specific to the team names, conferences, ORtg or DRtg column position, etc.

Extra Credit (Extra 1pt.)

- One of the most important observations in astronomy was made in 2018 with the coincident observation of gravitational waves in addition to photons across a wide range of wavelengths from a binary neutron star merger (https://arxiv.org/abs/1710.05833)
- There is an author list at http://www.nbi.dk/~koskinen/Teaching/AdvancedMethodsInAppliedStatistics2026/data/AuthorList.pdf
  - How many unique authors are there in the author list? (0.5 pts.)
    - Assume that all author entries are unique even if there seemingly are duplicates. E.g. my author name is "D. J. Koskinen" which only appears once. But if there were two instances of "D. J. Koskinen", then this should correspond to two different people.
  - If there was one single author list in alphabetical order (instead of being grouped by scientific collaboration), what author is the mid-point? (0.5 pts.)
    - What is author at the location (total authors)/2? Potentially there are two authors depending on whether the total number of authors is an odd or even number.
    - Sort by last name and then first initial(s).
    - Last names with first letters that include diacritics (e.g. Álvara) should go at the end of the alphabetical order.
    - Collaborations are not authors, e.g. "The 1M2H Team" should not considered as a unique author.